3D Reconstruction with Segment Anything Model (SAM) 3D

How artificial intelligence turns a single photograph into a complete 3D model

3D reconstruction — what is it? When you look at a photo of a chair, your brain immediately recognizes what the back looks like, even though it remains hidden. This ability to understand 3D shapes from a flat 2D image is natural for humans, but it has been a significant challenge for computer science for several decades. Previous methods required dozens of images from various angles or expensive equipment like laser scanners and depth sensors. The primary problem is that a single photograph completely lacks depth data, so the computer must analyze shadows, perspective, and surface textures to reconstruct what was lost when the image was captured as a flat picture.

HOW DOES IT WORK?

How 3D reconstruction works

From a flat photograph to a complete 3D model — five steps your brain does in an instant

Step 1

Flat photo
One ordinary image — no special camera needed.

Your phone is enough.

Step 2

Segmentation
AI outlines every object, separating it from the background.

Like scissors — but in milliseconds.

Step 3

Depth inference
Reads shadows and edges to estimate the distance of each part.

Your brain does this without thinking.

Step 4

Mesh building
Depth becomes a wireframe of thousands of triangles.

Like a scaffold before the walls go up.

Result

3D output
Shape, colour + depth. Rotatable & exportable.

Games · AR · medicine · design.

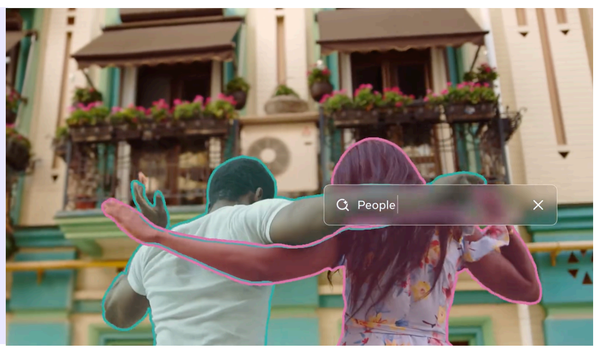
Medical applications – where 3D reconstruction could save lives

3D organ modeling from scans
Surgeons today use MRI or CT scans — flat 2D slices. 3D-style models can stitch these into full 3D organ reconstructions, helping surgeons plan operations more precisely and reduce risk.

Custom prosthetics and implants
A 3D model of a patient's limb or joint can be used to design precisely fitting prosthetics, reducing cost and production time dramatically.

Rehabilitation and movement analysis
Tracking a patient's posture and movement from a simple video. Physiotherapists can monitor recovery progress without expensive motion-capture equipment.

SAM 3D — how does it approach the problem? SAM 3D is an artificial intelligence model created by Meta AI Research that performs single-image 3D reconstruction. Instead of needing several camera angles or expensive sensors, the system was trained using a massive collection of photos and confirmed 3D scans. Through this process, it identified patterns regarding how common items and humans appear from any direction. By examining a single photo, the AI identifies details like lighting, outlines, surface textures, and object placement to generate a complete 3D mesh. This result is essentially a wireframe skeleton finished with the original colors and material properties.



YOUR TASK

1 Get started
Explore the iPad app

- **Form teams** — grab one iPad per team
- **Open Meta SAM 3D** — find it on the home screen
- **Explore the UI** — what options can you see?

2 Create 3D Scenes
From images

- **Open "Create 3D scenes"** — tap the menu option
- **Pick a sample image** — choose one from the gallery
- **Tap an object in the image** — select it with a single tap
- **Tap "Generate 3D"** — watch the magic happen ✨
- **Try your own images** — go back to home & explore

3 Create 3D Bodies
3D avatars of your team

- **Open "Create 3D bodies"** — tap the menu option
- **Pick a sample image** — choose one from the gallery
- **Tap a person in the image** — select them with a single tap
- **Tap "Generate 3D"** — create their 3D avatar ✨
- **Try your teammates!** — photograph & generate them 🎉

CAREERS THIS TECHNOLOGY CONNECTS TO

- Computer Vision Researcher
- Medical Imaging Scientist
- Biomedical Engineer
- 3D Artist / Technical Director
- Prosthetics Designer
- AR/VR Developer

SAM3D
Segment Anything in 3D

Want to try it at home?

Scan the code

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